GH3036 Wearing Detection Performance Report

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Preface

Purpose

This document is intended to reflect the Wearing detection performance test result of GH3036.

Audience

This document is intended for:

- Systems and product engineers
- Field application engineers
- Test engineers
- R&D engineers

Release Notes

This document is the frist official release; the corresponding software version is Package_V00.02.00.00

Revision History

Version	Date	Description
1.0	2025-11-14	The first release

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Contents

Preface I
1 Config Introduction 1
2 Test Requirements 2
3 Result 2
3.1 True Wear off Rate for objects (Summarized) 3
3.2 True Wear off Rate for objects (without interval) 3
3.3 True Wear off Rate for objects(with interval) 4
3.4 True Wear off Rate for objects(Pocket & Bag) 4
3.5 False Wear off Rate affects the skin 5
3.6 True Wear on Rate affects the skin 5
3.7 False Wear on Rate for objects 5
4 Terminology 5

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1 Config Introduction

Software version: Package_V00.02.00.00, mainly includes:

1. APP version: GHealth_1.0.9.13.apk
2. Algorithm version:

   GH_NADT_pre_pv_v1.0.2.0_nc
3. Firmware version: GR551x_VX.X_DEMO_DRV_LIB_GH3036.bin
4. Driver library version: GH(M)3X_DEMO_v0.2.1.0_ALGO.bin

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2 Test Requirements

Table 2-1 Test Requirement Details

Test Scope	Description
Test environment	Office room, walk-in constant temperature and humidity laboratory
Test environment temperature	25°C, -5°C
Testee gender	Male and female
Testee age	20- 40 years old
Testee BMI	18- 30 kg/m^2
Testee skin color level	Levels 2- 5
Test Distance	Watch on the surface of the object to be tested (without interval); Watch 2.5cm from the surface of the object to be tested (with interval)

Note:

Skin Tone Guide:

The Pantone SkinTone Guide										
Goodix Index	1	2	3	4	5	6	7	8	9	10
Pantone Index	1Y01SP	5Y01SP	2Y06SP	4Y07SP	2Y10SP	3Y11SP	2Y13SP	3R13SP	1R14SP	4R15SP

3 Result

1. Sample Number: Same as recorded in the table;
2. Procedure: Wear GH3036 properly (with moderate tightness to ensure no slipping during activity) with the wearing position at least 1 finger width up from the wrist bone. Then perform tests in different scenarios.
   - PPG
     1) True wear-on rate: Wear the watch properly and observe whether the wearing can be identified within 5 seconds;
     2) False wear-on rate: Place the watch on the surface of one object under test and observe whether the placement is identified as watch-wearing within 5 seconds;
     3) False wear-off rate: Wear the watch properly, rotate the wrist and shake the hand, to observe whether the wearing-off is identified within 1 minute; wear the watch loosely, rotate the wrist to make the watch contact with the test bench and observe whether the wearing-off can be identified within 1 minute;
     4) True wear-off Rate: Keep the watch worn-on for a period of time after test starts, switch the testing face which contacts the skin to contact an object instantly, and observe whether the wearing-off can be identified successfully within 1 minute;
     5) True wear-off time: Keep the watch worn-on for a period of time after test starts, switch the testing face which contacts the skin to contact an object instantly, and record the time required when the wearing-off is identified.

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3.1 True Wear off Rate for objects (Summarized)

Table 3-1 True Wear off Rate (Summarized)

Index / Scene	5s Wear off Rate	10s Wear off Rate	20s Wear off Rate	1min Wear off Rate	AVG Wear off Time(s)	# of scenes
Without Interval	60.61%	98.79%	100.00%	100.00%	3.34	330
With Interval	72.73%	97.73%	99.09%	100.00%	2.80	220
Pocket & Bag	77.50%	90.00%	93.75%	95.00%	3.54	80

3.2 True Wear off Rate for objects (without interval)

Table 3-2 True Wear off Rate (without interval)

Index / Scene	5s Wear off Rate	10s Wear off Rate	20s Wear off Rate	1min Wear off Rate	AVG Wear off Time(s)
Rough straw paper	10	10	10	10	1.00
Toilet paper	10	10	10	10	1.00
Blue book	10	10	10	10	1.00
Light-colored desktop	10	10	10	10	1.00
Wall	10	10	10	10	1.00
Aluminum can	10	10	10	10	1.00
Tempered glass screen protector	10	10	10	10	1.00
Cling film	10	10	10	10	1.00
Foam board	0	9	10	10	6.60
Washing machine	10	10	10	10	1.00
Dark desktop	10	10	10	10	1.00
Phone screen on	0	10	10	10	7.10
Phone screen off	0	9	10	10	8.70
Flat glass	10	10	10	10	1.00
Glass water glass	10	10	10	10	1.00
Frosted glass	10	10	10	10	1.00
Orange	0	10	10	10	6.40
Apple	0	10	10	10	6.00
Onion	0	10	10	10	6.40
Potato	0	10	10	10	6.00
Bed	10	10	10	10	1.00
Linen	0	10	10	10	6.20
Cotton cloth	0	9	10	10	7.30
Cotton-padded clothes	10	10	10	10	1.00
Umbrella	10	10	10	10	1.00
Wool sweater	10	10	10	10	1.00
Silk quilt	0	10	10	10	6.20
Red fabric	0	10	10	10	7.70
Gray fabric	10	10	10	10	1.00
Green fabric	0	9	10	10	8.00
Plush toy	0	10	10	10	7.70
Orange watch strap	10	10	10	10	1.00
Black watch strap	10	10	10	10	1.00

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3.3 True Wear off Rate for objects (with interval)

Table 3-3 True Wear off Rate (with interval)

Index / Scene	5s Wear off Rate	10s Wear off Rate	20s Wear off Rate	1min Wear off Rate	AVG Wear off Time(s)
Toilet paper	0	10	10	10	6.80
Blue book	10	10	10	10	1.00
Wall	10	10	10	10	1.00
Cling film	10	10	10	10	1.00
Tempered glass screen protector	10	10	10	10	1.00
Washing machine	10	10	10	10	1.00
Light-colored desktop	0	10	10	10	6.60
Dark-colored desktop	10	10	10	10	1.00
Mobile phone screen on	10	10	10	10	1.00
Mobile phone screen off	10	10	10	10	1.00
Flat glass	10	10	10	10	1.00
Bed	0	9	10	10	6.60
Linen	10	10	10	10	1.00
Cotton fabric	10	10	10	10	1.00
Cotton-padded coat	10	10	10	10	1.00
Umbrella	10	10	10	10	1.00
Wool sweater	10	10	10	10	1.00
Silk quilt	0	10	10	10	7.20
Red fabric	10	10	10	10	1.00
Gray fabric	0	10	10	10	6.00
Green fabric	10	10	10	10	1.00
Plump toy	0	6	8	10	12.50

Note:

In a scenario with gaps, two boxes of equal height are placed on the object under test, and the prototype is then raised 2cm off the ground.

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3.4 True Wear off Rate for objects (Pocket & Bag)

Table 3-4 True Wear off Rate (Pocket & Bag)

Index / Scene	5s Wear off Rate	10s Wear off Rate	20s Wear off Rate	1min Wear off Rate	AVG Wear off Time(s)
Placed in dark backpacks at rest	10	10	10	10	1.00
Placed in dark backpacks during exercise	10	10	10	10	1.00
Placed in light backpacks at rest	10	10	10	10	1.00
Placed in light backpacks during exercise	10	10	10	10	1.00
Placed in light pockets at rest	0	9	10	10	8.80
Placed in light pockets during exercise	2	3	5	6	13.50
Placed in dark pockets at rest	10	10	10	10	1.00
Placed in dark pockets during exercise	10	10	10	10	1.00

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3.5 False Wear off Rate affects the skin

Table 3-5 False Wear off Rate

Index Scene	False wear off Rate
Normal wearing demo	0/60
Loosen the wrist and rotate it	0/60

3.6 True Wear on Rate affects the skin

Table 3-6 True Wear on Rate

Index Scene	True Wear on Rate
Normal Wear	60/60

3.7 False Wear on Rate for objects

Table 3-7 False Wear on Rate

Index Scene	False Wear on Rate
Slowly contact various objects	0/140

4 Terminology

Table 5-1 Terminology

Indicator	Definition
True wear on rate	The ratio for system correctly recognizing wear-on status after users wear the device
False wear on rate	The ratio for system misrecognizing wear-on status after users take off the device
False wear off rate	The ratio for misrecognizing a wear-off event when users wear the device
True wear off rate	The ratio for system correctly recognizing wear-off status after users take off the device
True wear off time	The time for system correctly recognizing wear-off status after users take off the device
Wear on	The system recognizes the DUT is in wearing status.
Wear off	The system recognizes the DUT is in wear-off status.
Lead on	The system recognizes all the three electrodes of the DUT are in contact with human body.
Lead off	The system recognizes one of the three electrodes of the DUT is not in contact with human body.

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